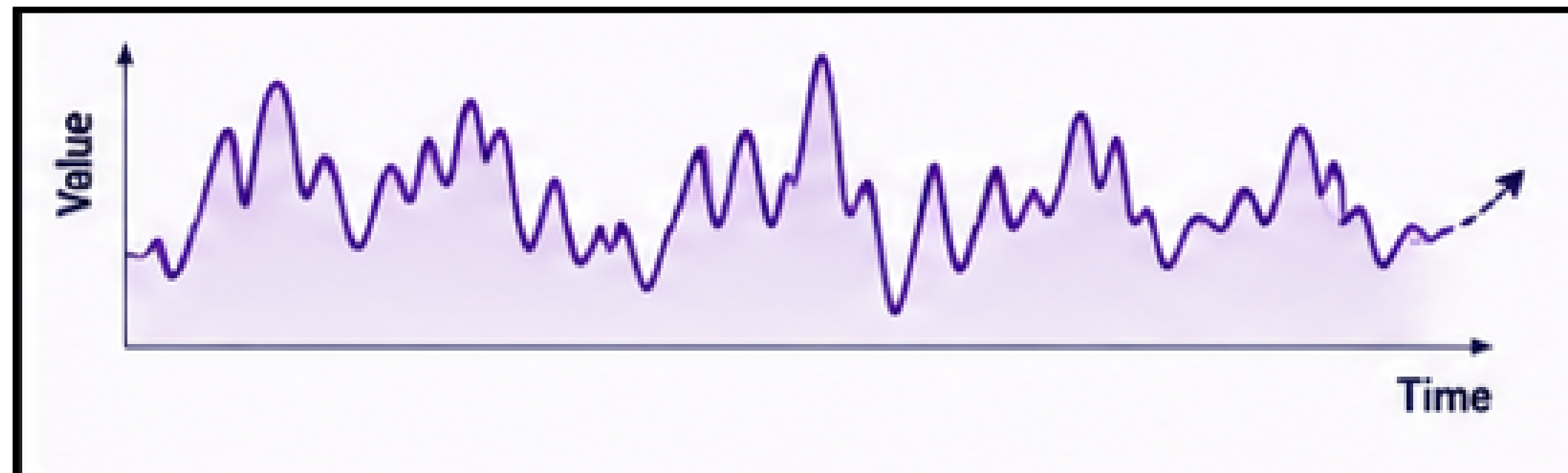


Do Similar patterns stay similar?

- That's the idea behind Sample Entropy (SampEn)
- It helps us understand how regular or unpredictable a time series is.

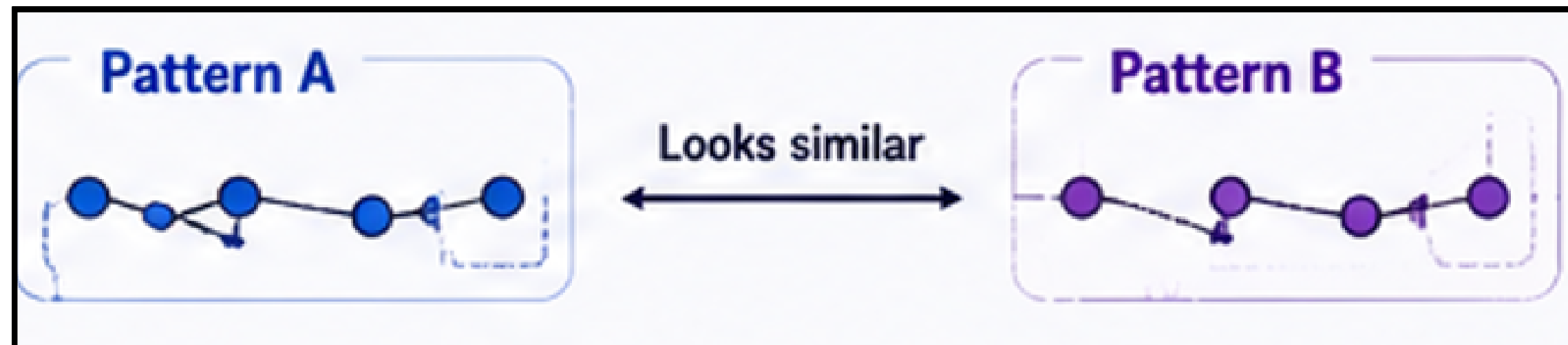


What is Sample Entropy?

- Sample Entropy = SampEn
It measures the regularity and complexity of a time series.
- If the pattern continues to behave similarly
 - **Low SampEn**
 - More regular
 - More predictable
- If the pattern continues to behave differently
 - **High SampEn**
 - More irregular
 - Less predictable

How does it work?

Imagine looking at short patterns in your data.



If it looks similar, then check if they continue to behave similarly or differently

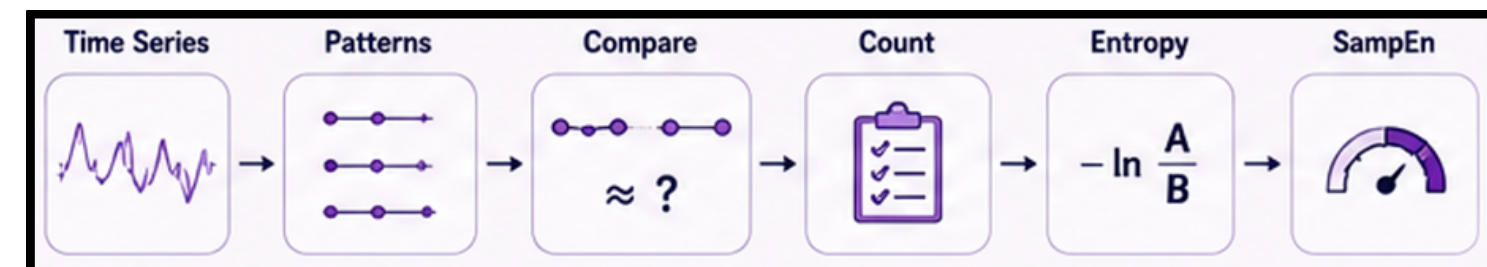


Similar pattern → Lower SampEn

Different pattern → Higher SampEn

How is SampEn Calculated?

- 1 Choose a pattern length m
- 2 Find similar patterns within a tolerance r
- 3 Compare whether those patterns remain similar when extended by one point
- 4 Calculate the ratio of these matches
- 5 Take the negative logarithm



$$SampEn(m, r) = -\ln \left(\frac{A}{B} \right)$$

where:

B = number of matching patterns of length m

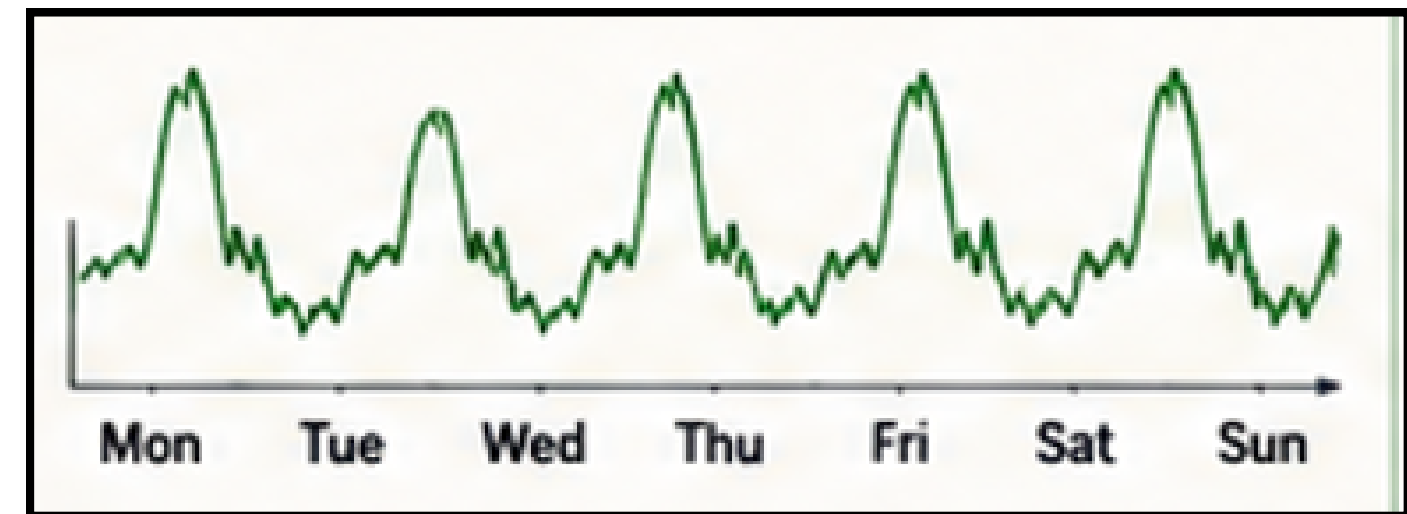
A = number of those matches that also remain similar at length $m+1$

Real-World Example

Imagine electricity demand ⚡

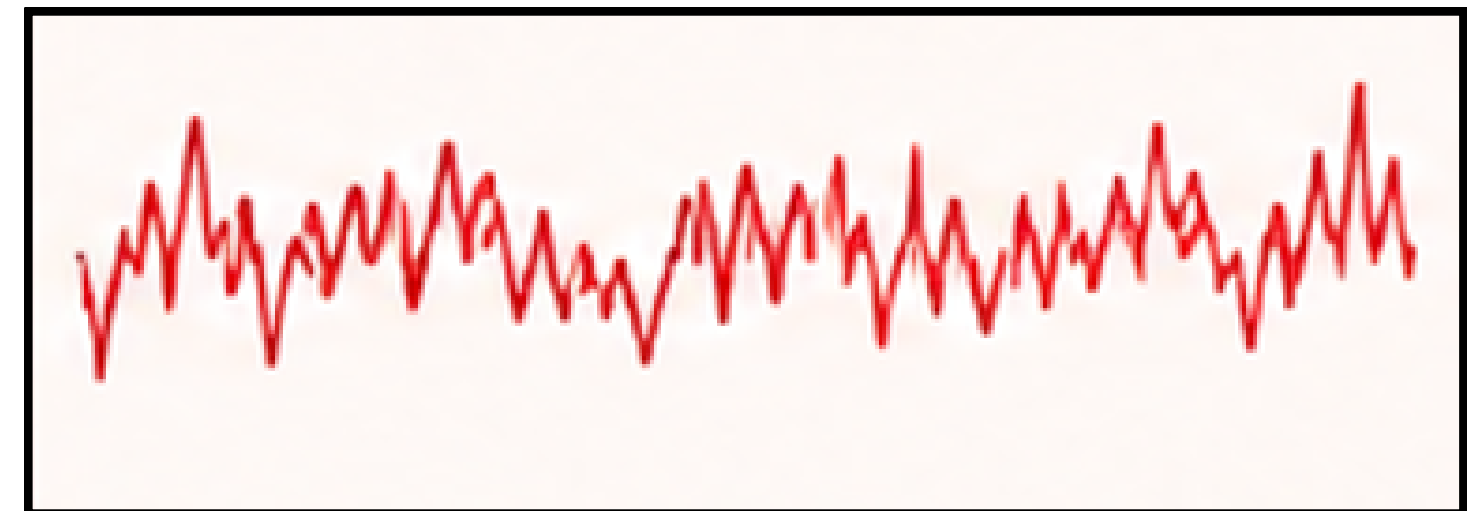
If similar daily patterns repeatedly continue in a similar way:

- More regular
- Lower SampEn
- Potentially more predictable



Now imagine a highly irregular signal:

- Similar patterns frequently break
- Higher SampEn
- Potentially less predictable



Key Takeaway

SampEn tells us about pattern consistency.

Low SampEn → Regular patterns

High SampEn → Irregular patterns